

Daniel Baxter

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Computer Science student at the University of Bath predicted a first class, with hands-on experience in competitive programming (ICPC) and cybersecurity CTF challenges. Comfortable working both independently and in agile teams through experience in an internship at Ditchcarbon and having developed the backend of a web app recognised as runner-up in two categories at Bath Hack 2026.

Education

Sep 2025 – June 2029

University of Bath

BSc (Hons) Computer Science with professional placement (Predicted First Class)

Year 1 modules (Average 84.5% to date):

- **Programming 1 (85%) & Programming 2:** Functional & object-oriented programming paradigms with algorithms, data structures and mathematical concepts. Individual work and documentation-focused group projects using Agile SCRUM methodology (Haskell, Java).
- **Discrete Mathematics and Databases (84%):** Set theory, predicate calculus with mathematical proofs. Relational models and mathematical foundations of databases (SQL).
- **Artificial Intelligence 1:** State-space search techniques, such as hill-climbing and adversarial searches. Probability theory, models for supervised and unsupervised learning. Individual projects in Python to implement constraint satisfaction & Naïve Bayes.
- **Computer Systems Architectures:** Computer architectures and hardware with operating systems. Components and registers of the CPU and low-level representations of data, including OS memory/file management.
- **Mathematics for Computation:** Linear algebra including matrices, Gaussian elimination and cryptography. Limits and calculus, Taylor/Maclaurin expansions.

Year 2 modules: Software Engineering, Algorithms & Complexity, Machine Learning, Visual Computing, Human-Computer Interaction 1&2, Advanced Programming, Cybersecurity.

Sep 2023 – June 2025

Northgate Sixth Form Ipswich, A Levels

Further Maths: A*, Maths: A, Computer Science: A, Physics: A.

Sep 2018 – June 2023

Northgate High School, Ipswich, GCSEs

11 GCSEs with nine grade 9s including Computer Science and Maths

Technical Skills

Programming languages: Python, C++, Haskell, Java, SQL

Frameworks/libraries: Dear ImGui (C++ GUIs), Django, Flask, PyTest, NumPy

Tools and technologies: CMake, Git, Godot game engine, Linux (Arch), Arduino

Projects

Web

- Collaboratively built the backend of a web app selected among runners up for the Best Design and Best Commercial Viability categories in the Bath Hack 2026 – ‘DropSpot’, a social media for leaving shared text notes across an interactive map (Available on GitHub).

Algorithms

- Developing a desktop GUI application in C++ for visually testing pathfinding algorithms on user-created graphs, using Dear ImGui and CMake (Available on GitHub).
- Independently implemented advanced data structures and algorithms in Python including Jump point search+ and Fibonacci Heaps, with a keen interest in algorithm optimisation.

Competitive Programming

- Completed competitive programming problems in Python with a team during the ICPC programming contest in October 2025.
- Completed cybersecurity challenges as a part of capture the flag events run by BAE Systems and CY2 involving exploiting bugs in C programs and vulnerabilities in web pages.

Other

- Researched and utilised finite state machines for character behaviour in the development of a small 2D action game using the Godot game engine.
- Worked in a small team advising the design of a mobile app 'Weirdswich' as part of a £1 million+ investment by the Ipswich Council, gaining insight into the research and advertising aspect of software.

Work Experience

June 2026 – September 2026 (upcoming): Internship, Ditchcarbon

Three month summer internship at Ditchcarbon; London-based, using AI to analyse and evaluate companies' Scope 3 emissions data and reduce carbon emissions. Involving learning full stack development in an agile startup environment.

June 2023 – June 2024

The Woolpack Pub, Ipswich: Pot washer/Kitchen Hand

13th June 2022 – 18th June 2022

IMT Aviation, Wendens Ambo: Work experience, aeroplane repair

February 2022 – September 2022

Suffolk Young Wardens: Volunteering to rebuild natural habitats around Suffolk, for DofE Bronze.

Interests/Activities

- Duke of Edinburgh Bronze, 2022
- Northgate Sixth Form Aspire Program 2024
- Badminton
- Chess
- Piano/guitar

Soft Skills

- Gained teamwork skills when rebuilding and enriching habitats for local wildlife in a group for the Suffolk Young Wardens as a part of the Duke of Edinburgh Bronze.
- Improved resilience and communication skills when working in a busy kitchen at the Woolpack Inn where time was key, and when planning a route as part of a team for the Duke of Edinburgh Bronze expedition.
- Learned self-motivation and organisation methods when planning my study, extracurricular and career plan for the Aspire Program at Northgate Sixth Form.

References: Available on request.